

Lecture notes from Winter 2025

Week **12** Lecture

Agenda

Types of Data

Reinforcement Learning

Ethics and Fairness

Future of AI

Week 12 - Announcements

Course Evaluations

- Please spare a moment to complete the course evaluations, it will help us guide the future development of the course.

It's quick.
It's confidential.
And it matters.

Fill out your course evaluations today.



Remaining Assessments

- (1) **Presentation** videos due [REDACTED]
 - Submit videos onto Quercus
- (2) **Final Report** [REDACTED]
 - Provide a short page/table summarizing individual contributions
- (3) **Final Exam** on [REDACTED]
 - Duration is 2.5 hours
 - EX200 (Exam Centre)

Presentations Schedule

➤ Presentations

➤ The presentations will be on Zoom:



➤ 2-4 minutes Q/A after each video, team members must be available to answer questions from audience (we want to learn from you!)

➤ Graded by **teaching team** (75%) and **classmates (25%)**

Presentation Details

- Your videos should be no more than 6 minutes
 - There will be up to 4 minutes of Q/A after video
- Try to focus more on your results – make it fun!
 - If your results are not great, put more emphasis on the qualitative findings
 - Emphasize your interesting examples
- Your video should have voiceover and can also include screen capture of PowerPoint.
- Be creative!

Project Report (Final Submission)

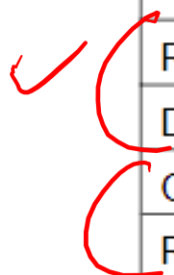
- Final project report is a continuation of the progress report – provides a **step-by-step walk-through of your project.**
 - brief project description supported with a diagram or illustration
 - data loading and any processing performed
 - model architectures
 - quantitative results (i.e., learning curves, accuracy, etc.)
 - qualitative results (i.e., interesting findings supported by examples/outputs from your model)
 - example of implementation/performance on new data
 - summary of related work (i.e., results on comparable projects)
 - brief discussion of the overall results. Do you think your model is performing well? Why or why not? What is unusual, surprising, or interesting about your results? What did you learn?

Project Guide (Final Submission)

Submission

- submit a .html (or .zip) file with everything required to run your code. You do not have to create a separate report.
- organized, **easy to read**, Jupyter notebook providing a step-by-step guide to completing the project.
- include a link to the code on GitHub or Google Collab.
 - share a link to your dataset or the original source of data

Grading

A red checkmark is drawn to the left of the first four rows of the table, and a red bracket is drawn around these same four rows, indicating that these components are the primary focus of the grading.

Component	Score	Weight
Results Achieved	10pt	2.5%
Discussion and Insights Gained	10pt	2.5%
Organization and Conciseness	10pt	2.5%
Reproducibility	10pt	2.5%
Total	40pt	10%

Last Minute Questions and Support

Please reach out to Instructor or Lead TA if you are running into any last-minute issues with the project.

Recap: Deep Learning

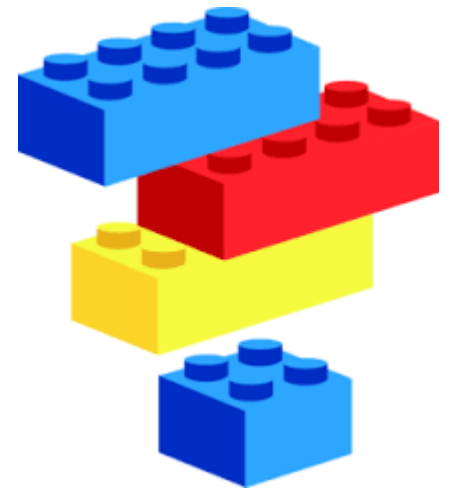
➤ Architectures

- linear layers, convolutional layers, convolutional transpose layers, recurrent layers, attention layers ...
- activation functions, skip connections, batch normalization ...

➤ Training

- loss functions, gradient descent, regularization, Adam optimizer, teacher forcing, sampling ...

➤ Data!!!!



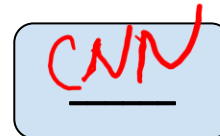
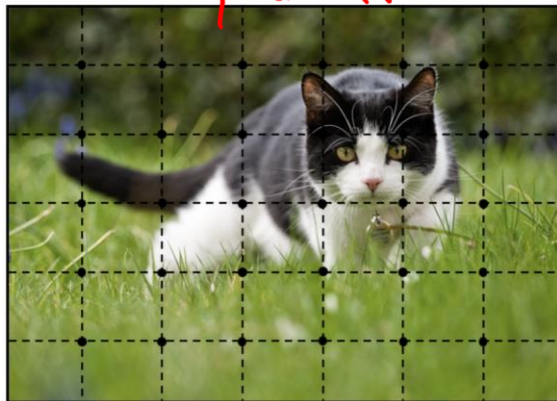
Many Forms of Data

tabular

age (n)	job (c)	marital (c)	education (c)	balance (n)	housing (c)
30	unemployed	married	primary	1787	no
33	services	married	secondary	4789	yes
35	management	single	tertiary	1350	yes
30	management	married	tertiary	1476	yes
59	blue-collar	married	secondary	0	yes
35	management	single	tertiary	747	no

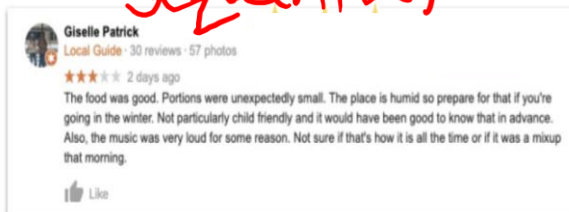


Income | \$54,000



Cat |

Dog |



|

|

|

|

Generating Data



Autoencoder
GAN



Autoencoder
GAN



to be or not to be



(decoder)
RNN
Transformer



That's the question.

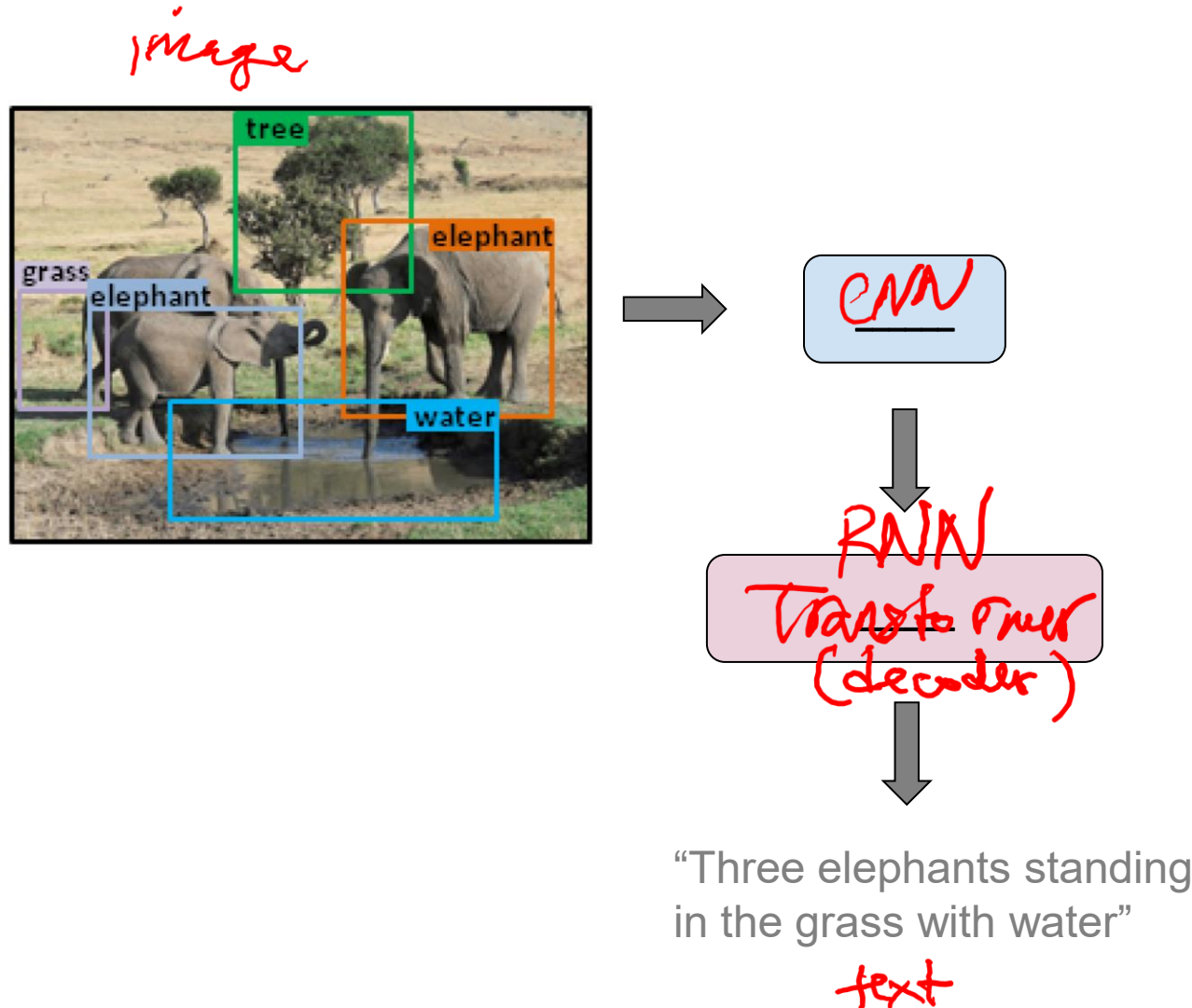
Most of us are kinda - sorta afraid to admit it.

Not having a clue on how or what to say.

So we stay silent, walking around in the dark, hurting, and scared.

We are afraid

Combining Models

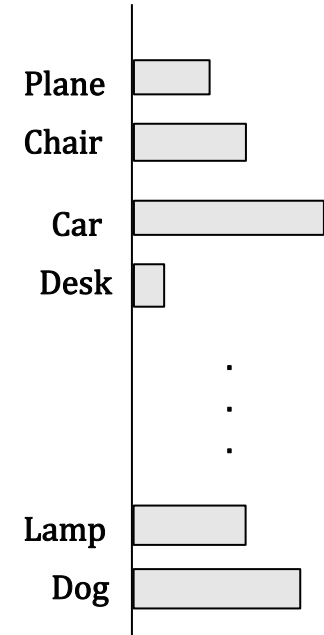


Point Cloud Data



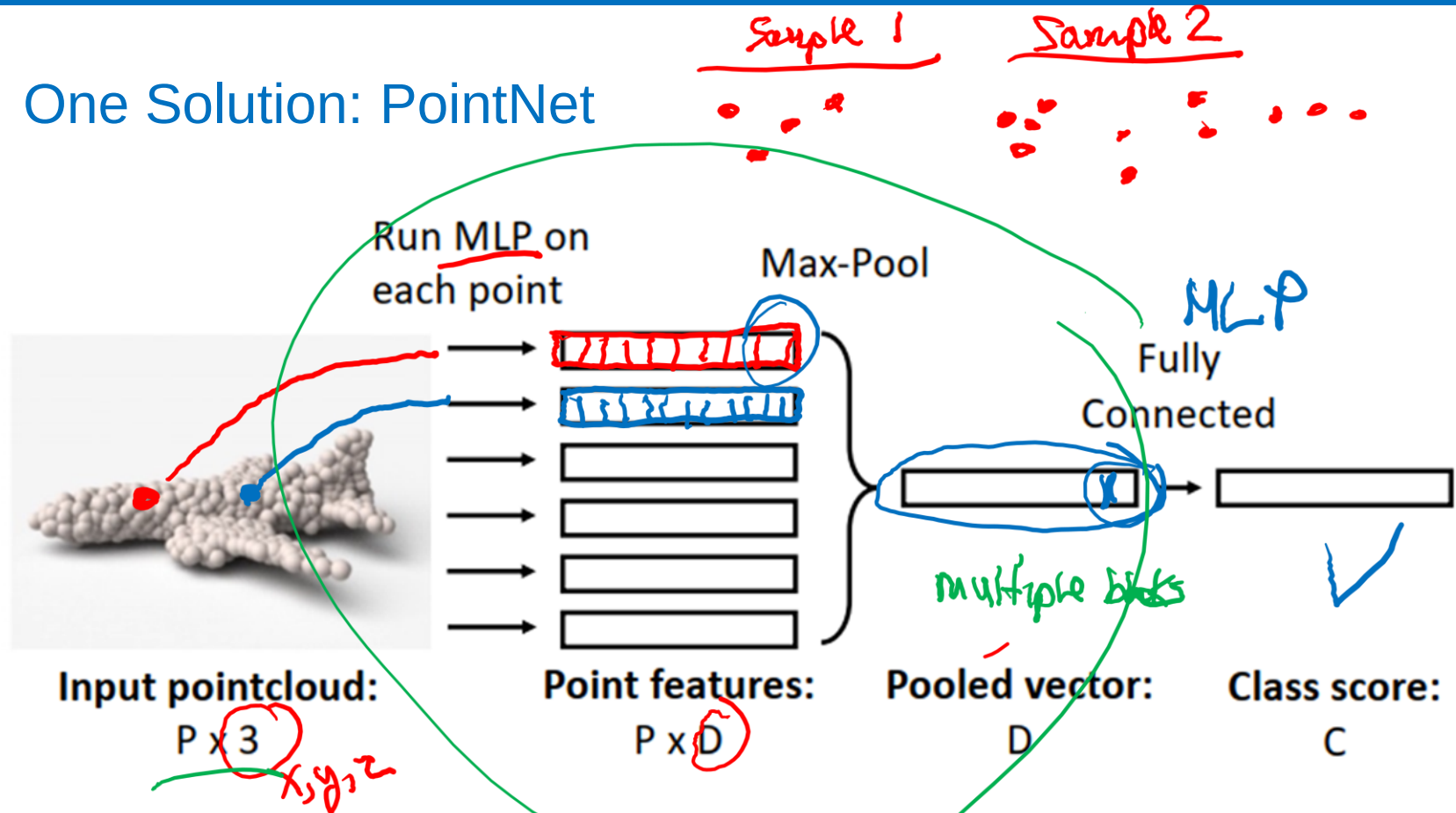
Q: What is special about point cloud data?

unordered points



Point Cloud Classification

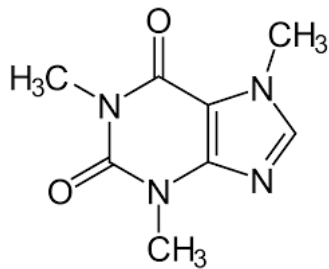
One Solution: PointNet



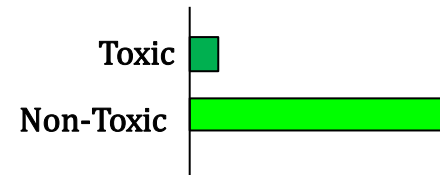
Qi et al, "PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation", CVPR 2017
Qi et al, "PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space", NeurIPS 2017

Graph Data

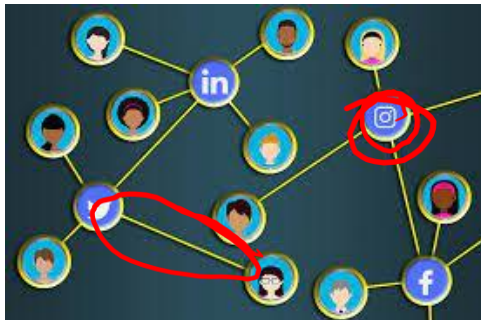
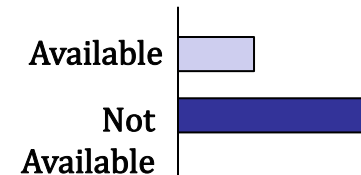
unordered data



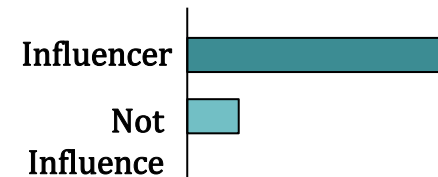
GNN



GNN



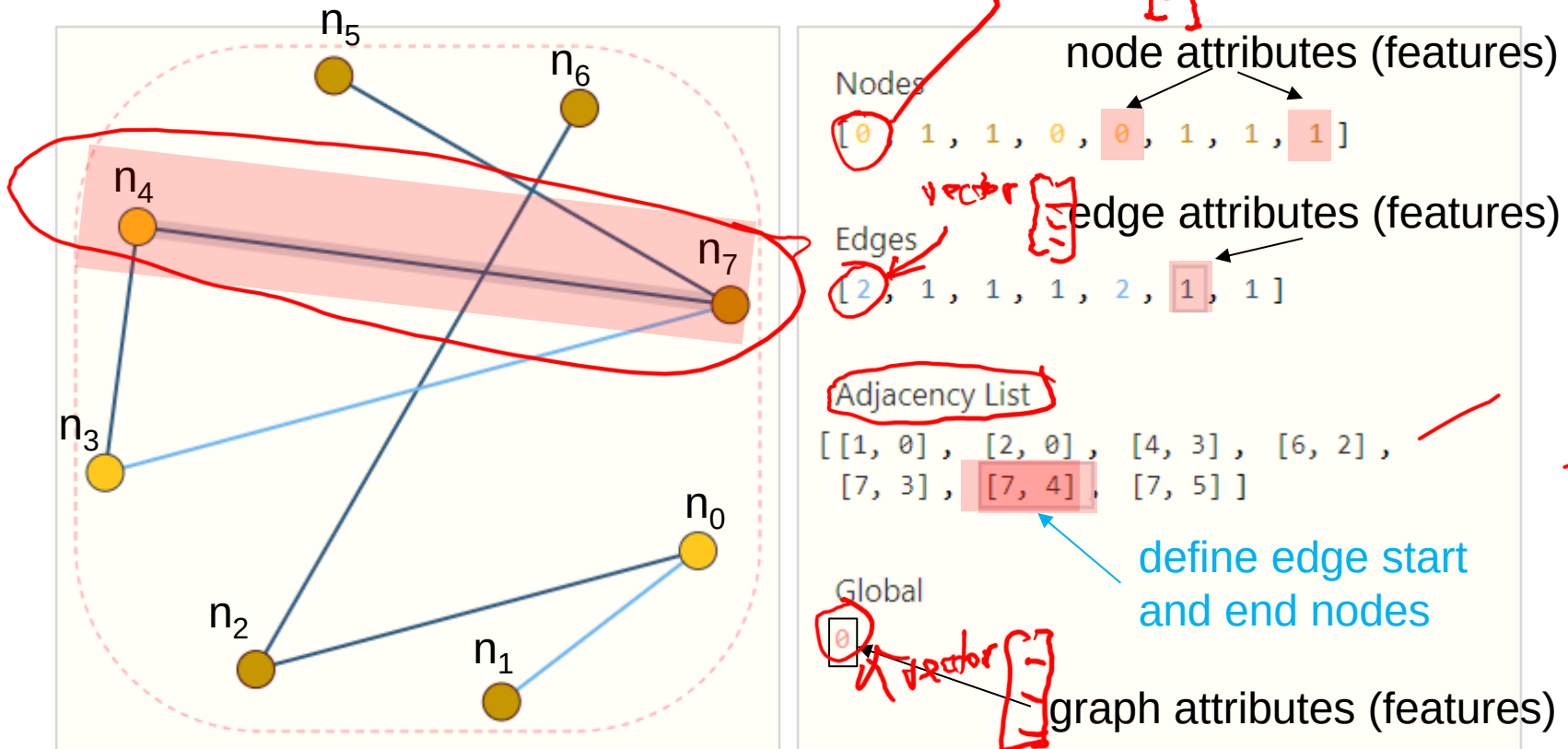
GNN



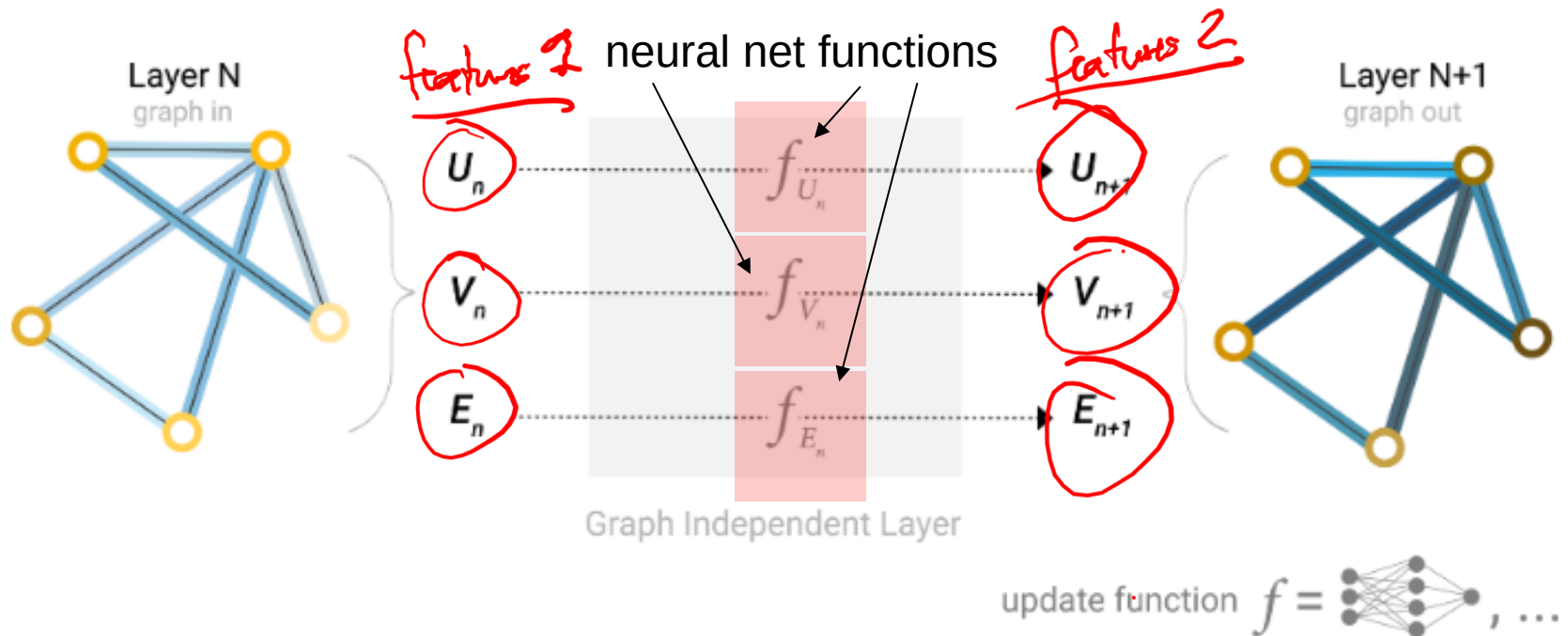
—

Graph Neural Networks

- GNNs can work with complex data structures



Graph Neural Networks



- V** Vertex (or node) attributes
e.g., node identity, number of neighbors
- E** Edge (or link) attributes and directions
e.g., edge identity, edge weight
- U** Global (or master node) attributes
e.g., number of nodes, longest path

There are many
different variants
of Graph Neural
Networks

Graph Neural Networks

- Many machine learning problems can be formulated into a graph network.

Images

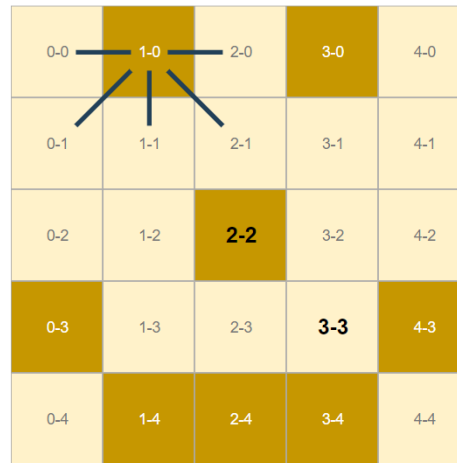
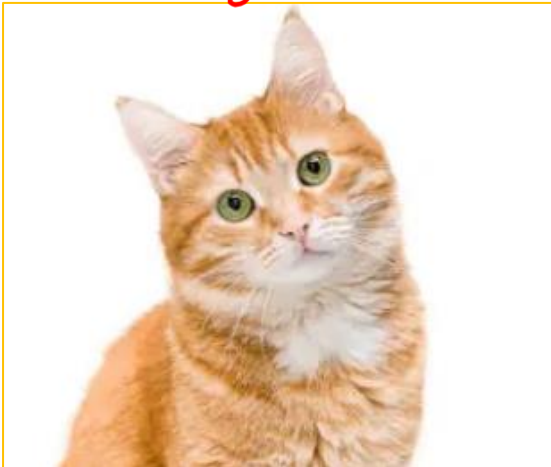
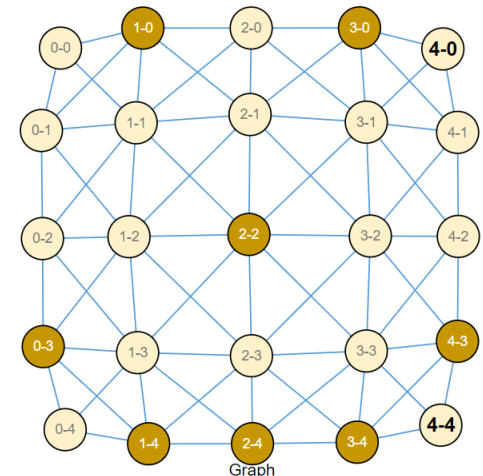


Image Pixels



Graph

text



Source: [distill](#)

GNNs are powerful architectures, but can be complicated to set up.

Deep Learning Paradigms

➤ **Supervised Learning**

- e.g., classification, regression, key points, pose estimation ...

➤ **Unsupervised Learning**

- e.g., generate images, image colourization, language model ...

➤ **Self-Supervised Learning**

➤ **Semi-Supervised Learning**

- e.g., image reconstruction, word embeddings ...

one more
important
learning
paradigm!

Continue at 11:10am

Reinforcement Learning

Motivation

How can we automate the steering of a car?



➤ Q: Can we use supervised learning to play a car racing game?

Challenges with Supervised Learning

- **Problem:** We don't have the "correct" answer (ground truth move) at each step
 - We have a signal to tell us whether our agent completed the game.
 - . . . but the signal is delayed . . .
 - we don't see the signal until the end of the game.
- Q: Why can't we just use win/loss information (at the end of the game) as a label for whether the moves were good/bad?

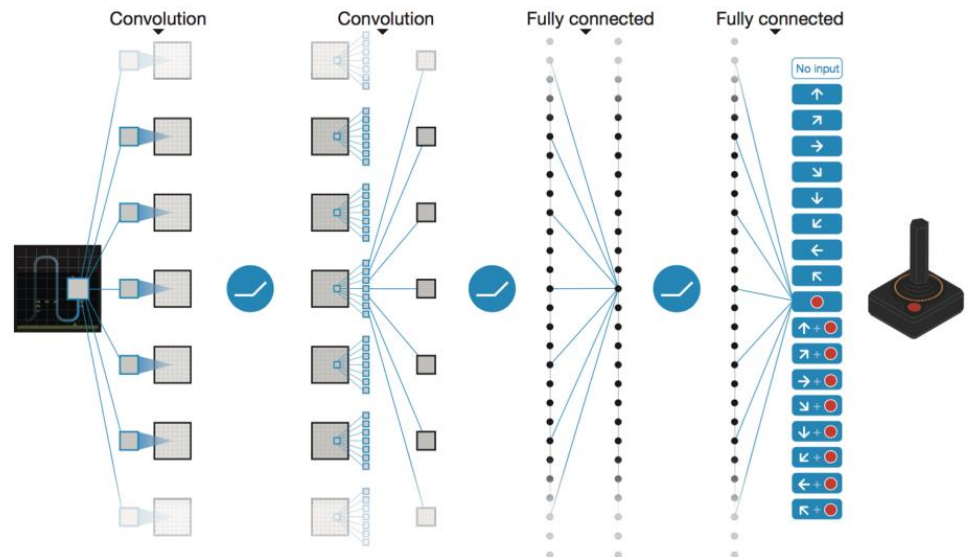
Obtaining Labels

- The more time has passed since we made our action the less accurate our labels will be.
 - credit assignment problem
 - e.g., waiting an hour to give a treat (reward) to your dog after it followed your command would not be effective in reinforcing positive behaviour.
- Q: Is there some way we could obtain more accurate labels for the sequences leading to a positive/negative (win/loss) result?

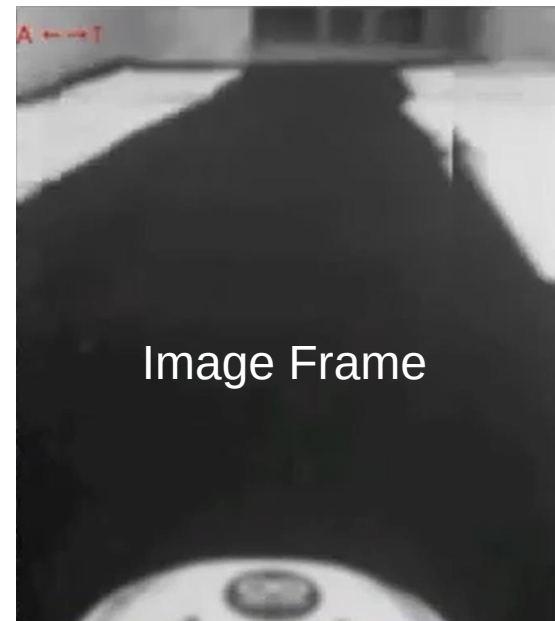
Obtaining Labels Con't

- **One solution:** make our own data with labels
 - e.g., in the case of a video game, we could play the game and **collect screenshots** along with **actions taken**.
 - then train a CNN to **predict the correct action** based on image data.

Known as
Imitation
Learning



Sample Video

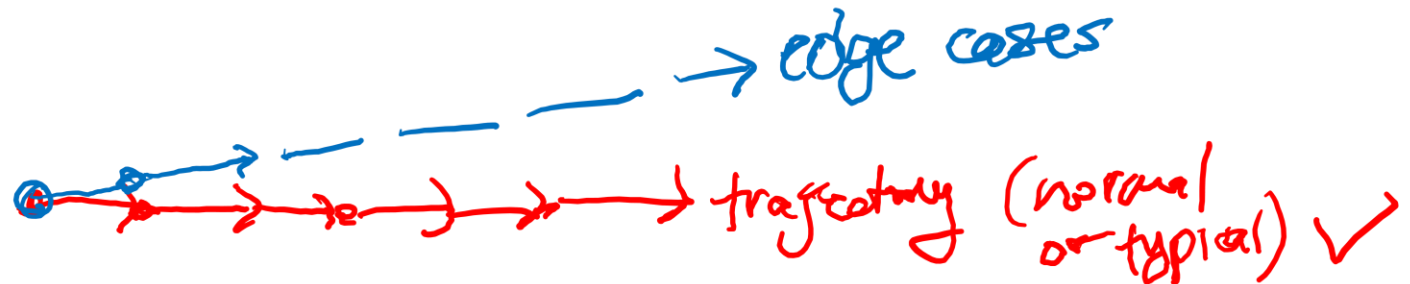


Limitations of Imitation Learning

Q: What are some limitations resulting from learning to imitate a human player?

1. limited to skills of the human - may not be the optimal

2.



Reinforcement Learning

- **Another solution:** use reinforcement learning
- Different from supervised learning:
 - there is no supervision (no “correct” answer), only a scalar **reward** signal
 - there is a notion of “time” (steps/moves)
 - feedback/reward signal is **delayed**
 - **nonstationary** - our decisions/actions will affect the subsequent data we receive

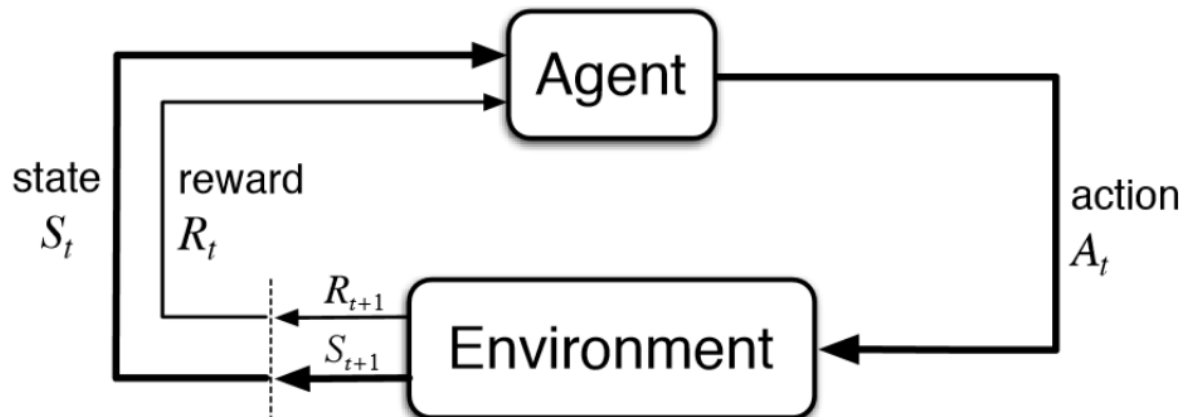
Reinforcement Learning Con't

➤ Environment

- Provides the agent with the current state
- Provides a reward at each time step (mostly 0 reward)

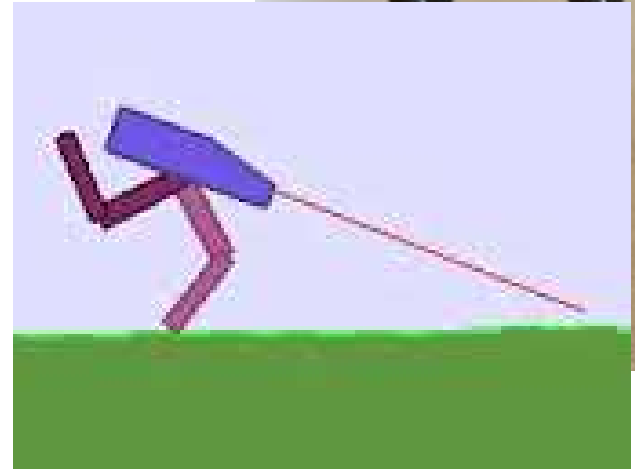
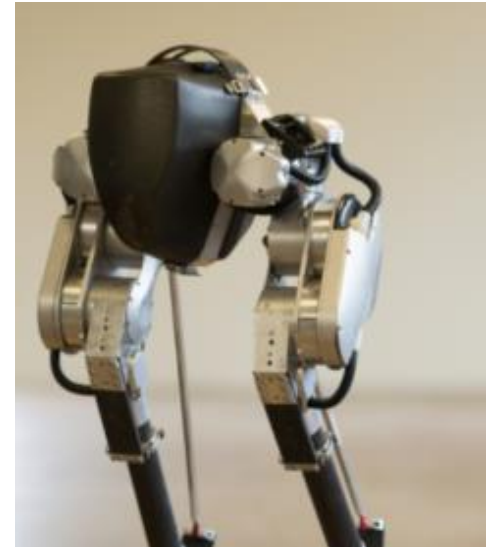
➤ Agent

- Autonomous entity that chooses an action at each time step, given the state
- Representing states and actions can be tricky



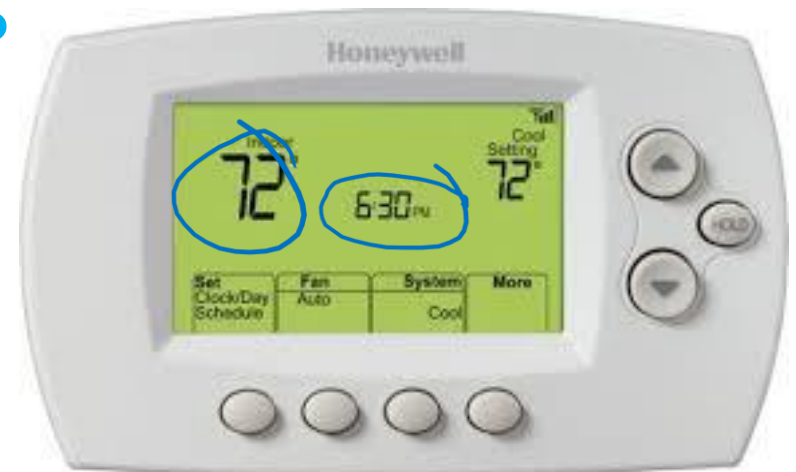
Example 1: Teach Robot to Walk

- Given a robot that has the ability to walk and has access to distance and touch sensors.
- How would we set up the system to teach the robot to walk?
- Q: What are the states?
- Q: What are the actions?

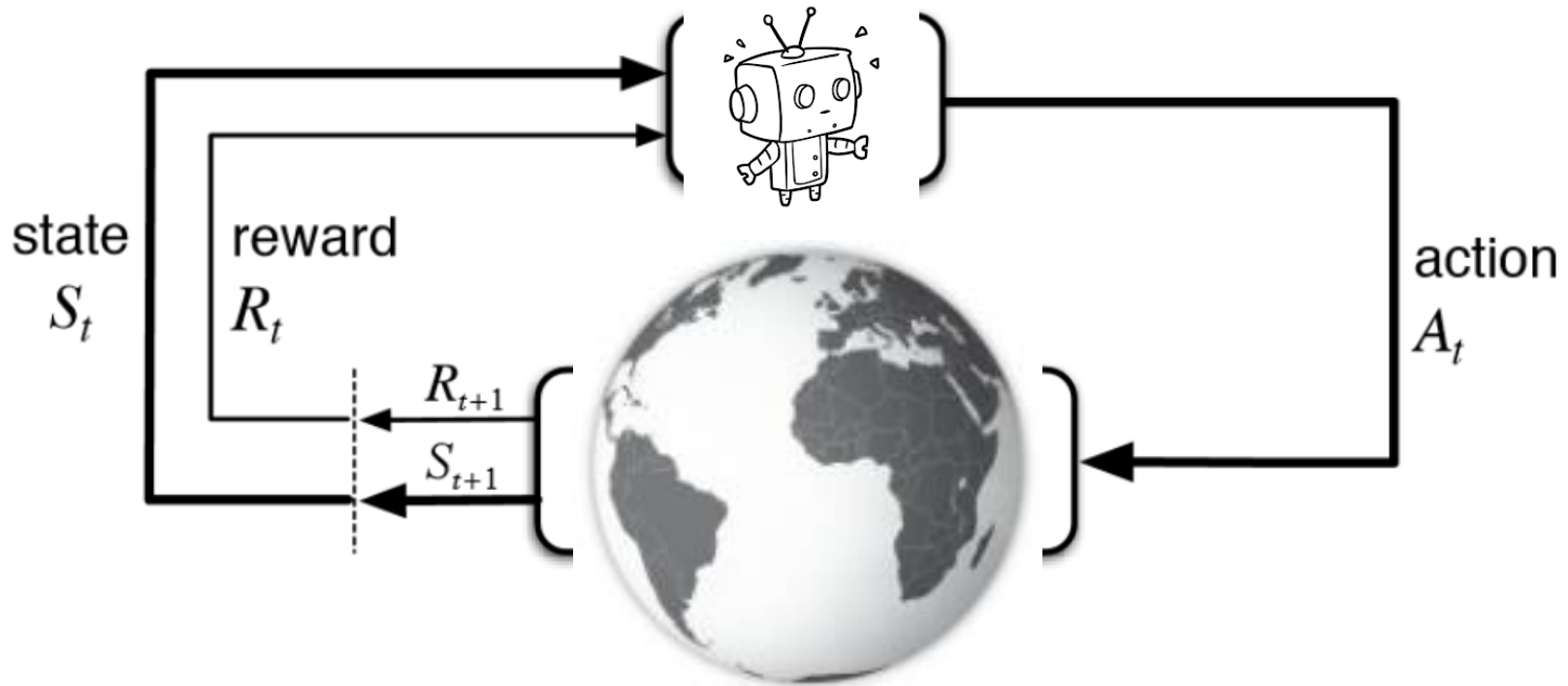


Example 2: Smart Thermostat

- Design a program to control a smart thermostat to anticipate the needs of the owners.
- Q: What are the states?
- Q: What are the actions?



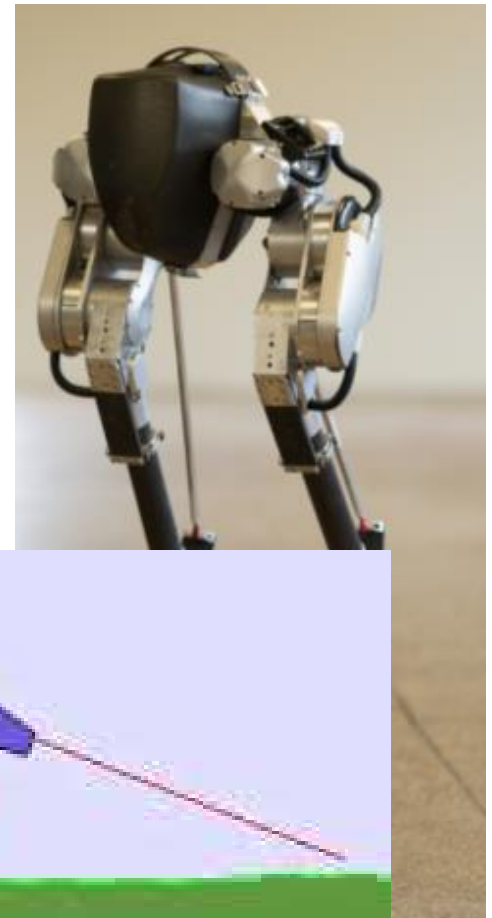
Reinforcement Learning Setup



- A **reward** R_t is a scalar feedback signal
- The goal of the agent is to select actions to maximize total discounted future reward.

Example 1: Teach Robot to Walk Con't

- Given a robot that has the ability to walk and has access to distance and touch sensors.
- Q: What are the rewards?



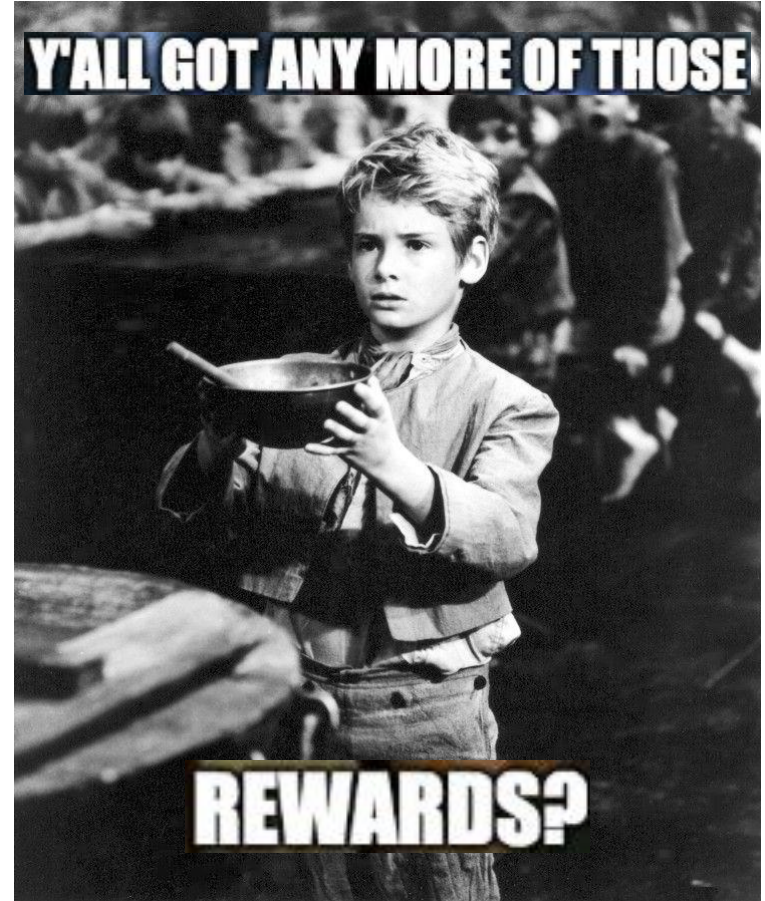
Example 2: Smart Thermostat Con't

- Design a program to control a smart thermostat to anticipate the needs of the owners.
- Q: What are the rewards?



Bad Rewards

- Setting up rewards to get good agent behaviour is tricky.
- If the reward is not well designed, it can lead to reward hacking



Example: Bad Rewards



- Instead of finishing the race, the agent finds a loophole in the game

<https://openai.com/blog/faulty-reward-functions/>

Classical Approach (Without Neural Nets)

Markov Decision Process

- Based on the early 20th century work by Andrey Markov and later refined by Richard Bellman in 1950s
- Mathematical formalization of the reinforcement learning problem using:
 - set of possible states
 - set of possible actions
 - distribution of rewards given (state, action) pair
 - distribution of transition probabilities over next state

tabular methods

RL Terminology

- **Agent:** autonomous entity (i.e., robot) that interacts with an environment
- **Policy:** an agent's behaviour function
- **Value function:** expected future **return** that can be obtained at the state given by applying the policy
- **Q function:** expected future **return** for all (state, action) pairs

Optimal Policy

- Given optimal Q function values, it is trivial to obtain the optimal policy $\pi(s)$:

$$\pi(s) = \text{argmax}_a Q(s, a)$$

↖ actions to take



Q (s,a)		Actions			
		→	↑	↓	←
States	0	0.2	0.3	0.1	0.9
	1	0.9	0.1	0.1	0.2
	2	0.3	0.3	0.4	0.1
	3	0.5	0.2	0.6	0.7
	4	0.1	0.2	0.1	0.1

Bellman Equations

- Estimate for the optimal state value, $V(s)$
 - sum of all future discounted rewards the agent can expect to receive in state 's'.

- Bellman equation:

$$Q(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s')]$$

$$V(s) = \max_a Q(s, a)$$

$T(s, a, s')$ is the transition probability.

$R(s, a, s')$ is the immediate reward.

γ is the discount rate

Discounted Return

- The **discounted return** at a time step G_t is defined to be:

$$G_t = \sum_{k=0}^T \gamma^k R_{t+k+1}$$

- Where $0 < \gamma \leq 1$ is a constant **discount factor**.
- With discounting, **getting a reward now is better than getting the same reward later**.

Value Iteration Algorithm

➤ Algorithm:

1. For all s set $V(s) = 0$
2. Repeat until convergence:
 - a. For all actions, a , determine and set $Q(s,a)$
 - b. $V(s) = \max_a Q(s,a)$
3. Return Q


$$Q(s,a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma \cdot V(s')]$$

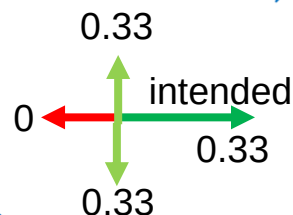
Example Problem

➤ Frozen Lake Example

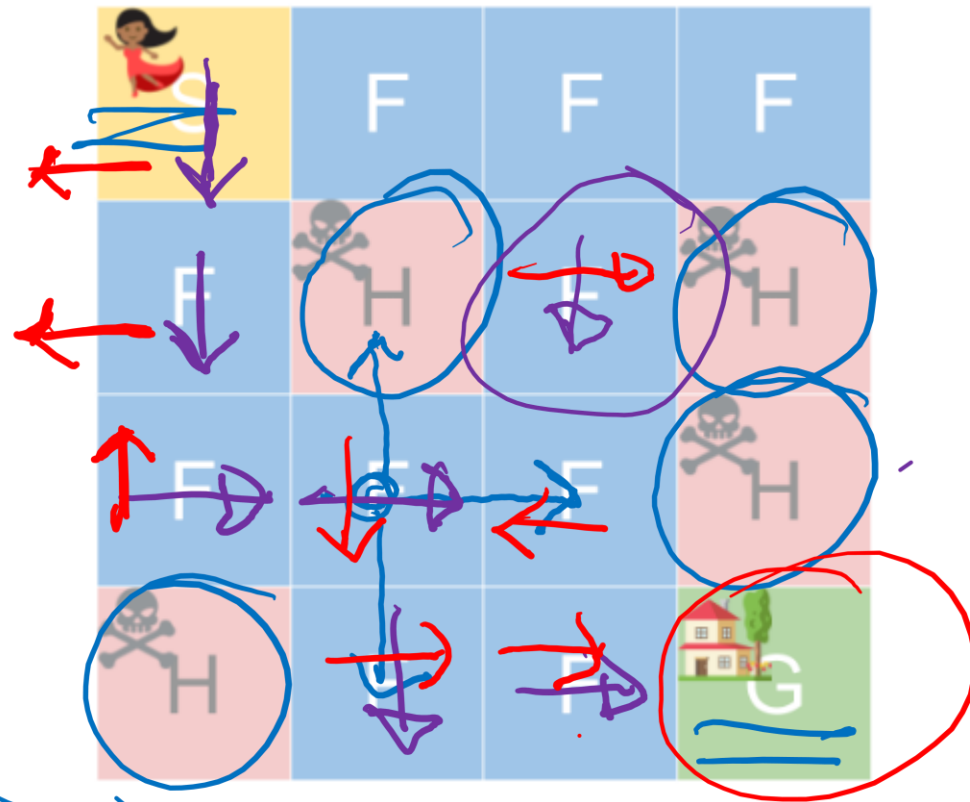
0:S 1:F 2:F 3:F
4:F 5:H 6:F 7:H
8:F 9:F 10:F 11:H
12:H 13:F 14:F 15:G

S starting location
F frozen location
H hole
G goal location

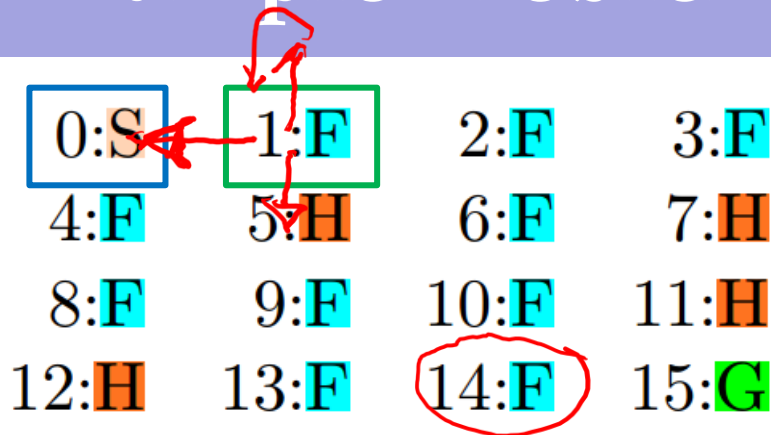
Transitions: Pr. opposite of
your intended direction is 0,
all other directions are 0.33



H => Reward = 0
G => Reward = 1



Example Problem Con't



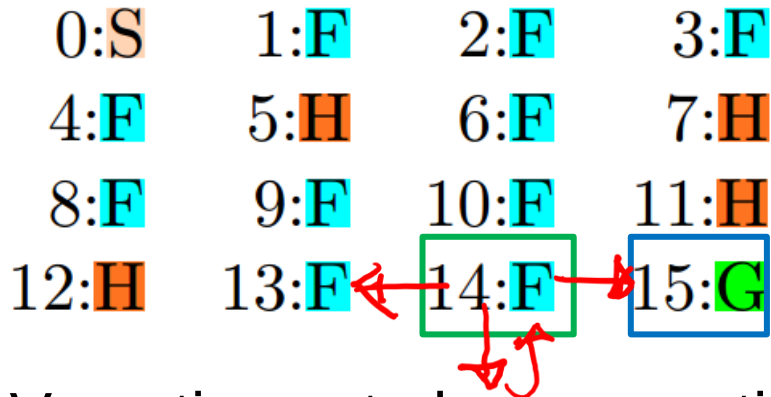
$$Q(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma \cdot V(s')]$$

Consider state 1. This requires computing $Q(1, a)$ for all four actions and then setting $V(1)$ to the maximum of the four Q values. Let's start with moving left:

$$\begin{aligned}
 Q(1, l) &= \overset{\text{State } = 0}{.33 \cdot (0 + .9 \cdot 0)} + \overset{\text{State } = 1}{.33 \cdot (0 + .9 \cdot 0)} + \overset{\text{State } = 5}{.33 \cdot (0 + .9 \cdot 0)} \\
 &= 0 + 0 + 0 \\
 &= 0
 \end{aligned}$$

Similarly for $Q(1, r)$, $Q(1, u)$, $Q(1, d) = 0$

Example Problem Con't



$$Q(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma \cdot V(s')]$$

V continues to be zero until we get to state 14, where we finally get nonzero values for $Q(14; d)$, $Q(14; r)$, and $Q(14; u)$:

$$\begin{aligned} Q(14, d) &= \text{state 13} \quad .33 \cdot (0 + .9 \cdot 0) + \text{state 14} \quad .33 \cdot (0 + .9 \cdot 0) + \text{state 15} \quad .33 \cdot (1 + .9 \cdot 0) = .33 \\ Q(14, r) &= .33 \cdot (0 + .9 \cdot 0) + .33 \cdot (0 + .9 \cdot 0) + .33 \cdot (1 + .9 \cdot 0) = .33 \\ Q(14, u) &= .33 \cdot (0 + .9 \cdot 0) + .33 \cdot (0 + .9 \cdot 0) + .33 \cdot (1 + .9 \cdot 0) = .33 \\ Q(14, l) &= ? \approx 0 \end{aligned}$$

Hence, $V(14) = 0.33$

Example Problem Con't

0:S 1:F 2:F 3:F
4:F 5:H 6:F 7:H
8:F 9:F 10:F 11:H
12:H 13:F 14:F 15:G

On iteration two most values stay 0, However, this time states 10 and 13 also get nonzero Q and V entries since we can get to state 14 from them and from the last iteration $V(14) = .33$.

$V(s)$
4x4
rep.

0	0	0	0
0	0	0	0
0	0	0	0
0	0	.33	0

After iteration 1

0	0	0	0
0	0	0	0
0	0	.1	0
0	.1	.46	0

After iteration 2

Let's look at some code!

Q-Learning

- Typically:
 - The agent **does not** have information on the state transition probabilities $T(s,a,s')$ *don't have those*
 - ... and the agent **does not know** what the immediate rewards $R(s,a,s')$ are going to be.
- We can modify our Value Iteration Algorithm into what is called a **Q-Learning Algorithm**.

Q-Learning Algorithm

➤ One possible algorithm:

1. For all s set $Q(s,a) = 0$, and $V(s) = 0$

2. Repeat until convergence:

Loop
a. Randomly select state “ s ” and randomly select action “ a ” from available actions at state “ s ”

b. If immediate reward available update matrix $R(s,a)$

c. Update $Q(s,a)$

d. $V(s) = \max_a Q(s,a)$

3. Return Q

$$Q(s,a) = (1 - \underbrace{\alpha}_{\text{old info.}}) Q(s,a) + \underbrace{\alpha}_{\text{new info.}} [R(s, a, s') + \gamma V(s')]$$

Exploration vs Exploitation

- **Exploitation**: Take actions the function already thinks will lead to a good outcome
- **Exploration**: Try making novel actions and see if you discover a way to adjust the function to get even better outcomes
- Need a good **balance of exploration vs exploitation** to learn a good RL agent

Epsilon Greedy Policy

- Idea is to learn and wander at the same time, and allow the learning to influence where we go
- Use a “greedy” approach
 - $1 - \epsilon$ probability to choose action with the highest estimated return
 - ϵ probability of to choose a random action
- Q: How should we change ϵ over time?

Note:
should apply this
when training text
generative
models using
teacher forcing!

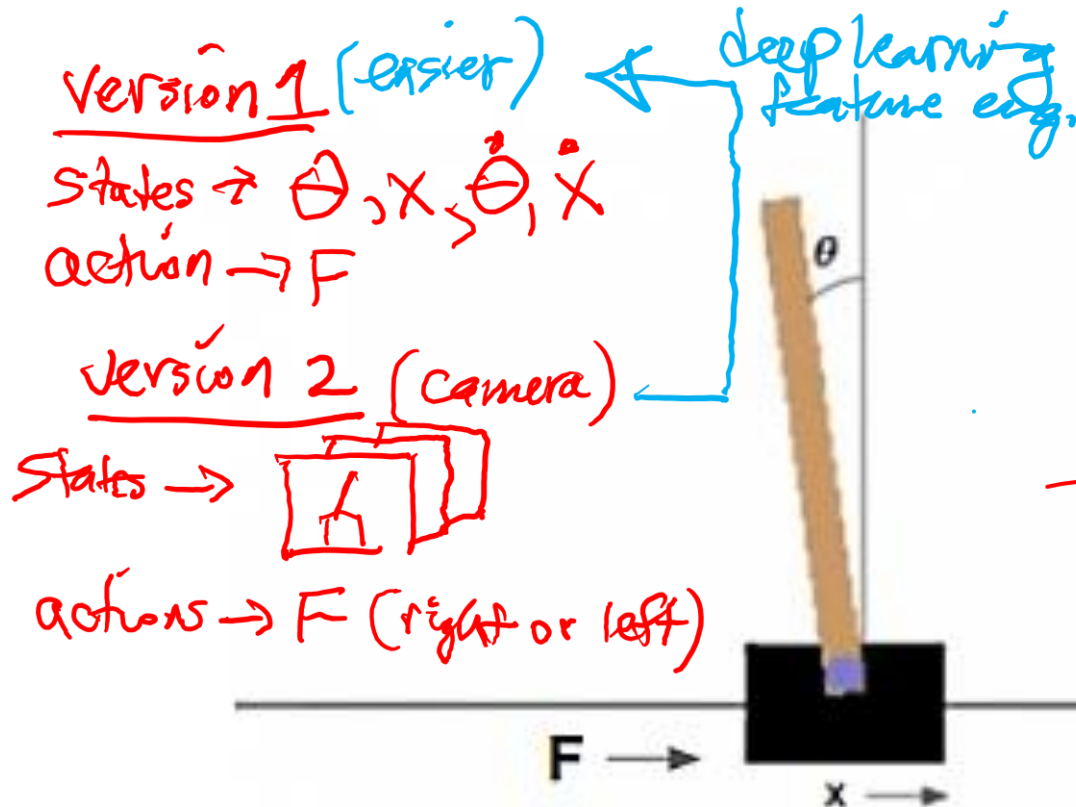
Let's look at some code!

continue at 12:18pm

“Deep” Reinforcement Learning

Inverted Pendulum Problem

- **Goal:** keep the pendulum from falling and going off screen



➤ Q: What are our states?

➤ Q: What are our action?

Limitations of Tabular Methods

- Q: Why would the classical (tabular) reinforcement learning approaches fail on the inverted pendulum problem?

data is continuous (states) $\rightarrow$ infinite possibilities
does not fit into a table

- Q: How can neural networks help?

use neural network $\Rightarrow f(x)$
to model $Q(s, a)$

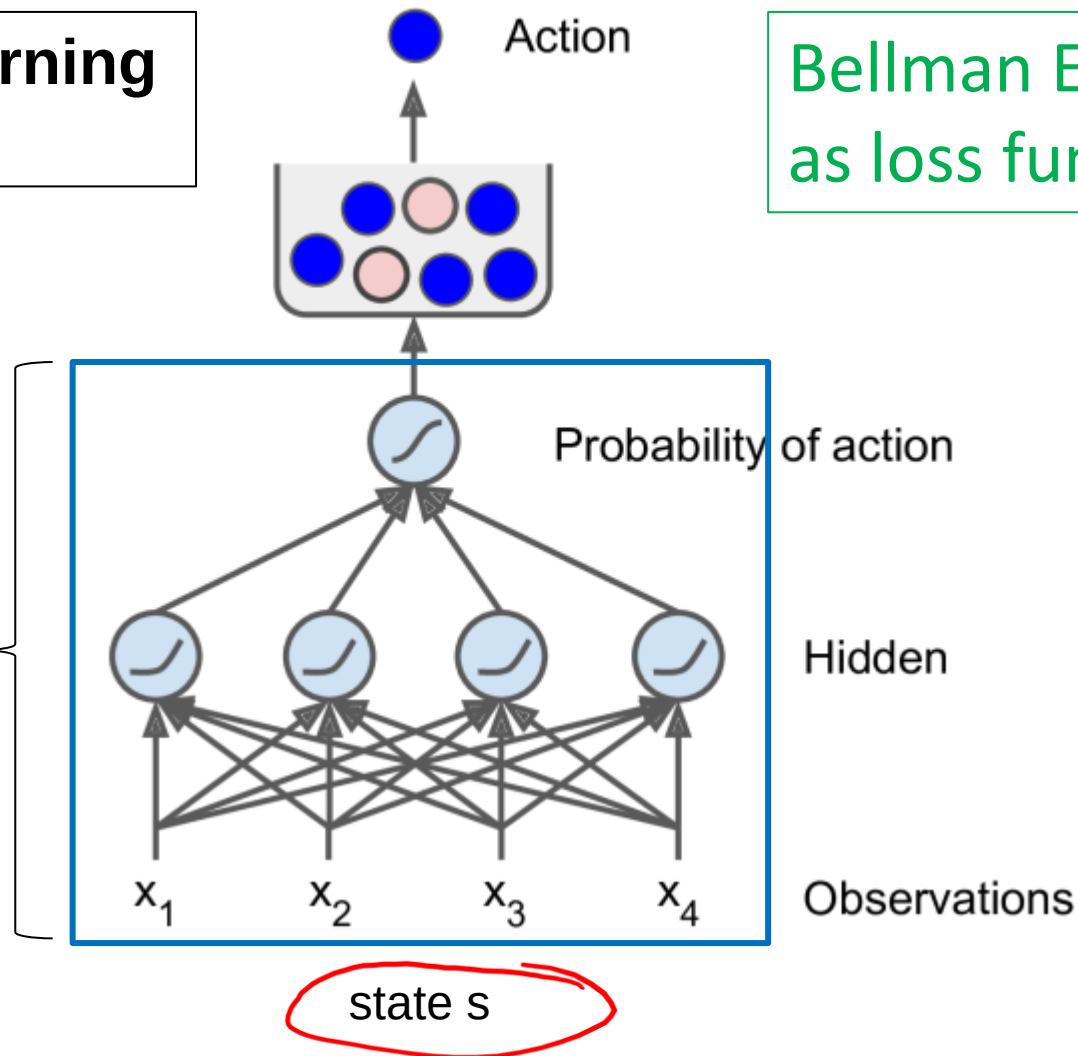
Deep Learning to the Rescue!

Deep Q-Learning (DQN)

Bellman Equation as loss function!

Key concept:
Neural Networks
can approximate
functions

$Q(s,a)$



DQN Loss Function

- Temporal Difference (TD) Loss Function:
 - difference between the current Q-value and the target Q-value (reward plus the discounted maximum future Q-value).

$$\boxed{Q(s, a)} - (\underbrace{R(s, a)} + \underbrace{\gamma}_{\text{when converges}} \max_{a'} Q(s', a'))^2 = 0$$

- It can be computed after each step or action and the difference should converge to zero with training

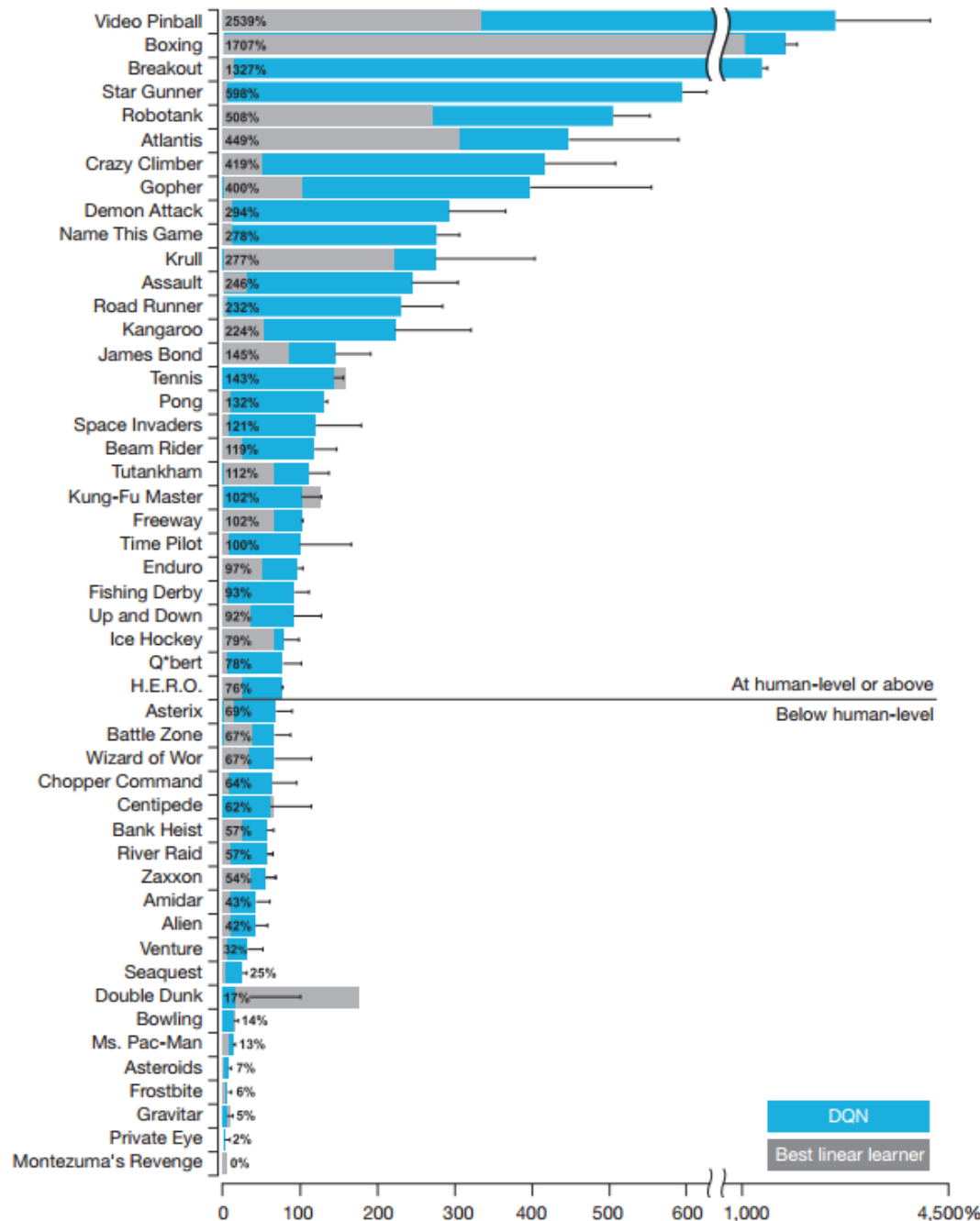
Deep Q-Learning

- Developed in 2013 by Vlad Mnih et al. at DeepMind
(former Geoffrey Hinton's grad. student)
- Achieved state-of-the-art results on Atari games,
and **surpassed human-level performance**
- Algorithm **generalized** over other Atari games
 - Prior to this algorithms required extensive tuning
for each game
 - **Artificial (general) intelligence**
- DeepMind was purchased by Google in 2014

Figure 3 | Comparison of the DQN agent with the best reinforcement learning methods in the literature.

The performance of DQN is normalized with respect to a professional human games tester (that is, 100% level)

Performs at a level that is broadly comparable with or superior to a professional human games in the majority of games.



Source: Mnih et al., (2013)

Many Approaches to RL

➤ Main Deep Learning Approaches:

- Q-Learning (learn Q function)
- Policy Gradients (learn distribution of actions for given state)

➤ Other Approaches:

- Actor-critic
- Model-based
- Adversarial
- ...

Reinforcement Learning
is a big topic that can be
taught over several
course (e.g., APS1080)

Training ChatGPT = GPT + RL + HF

Step 1

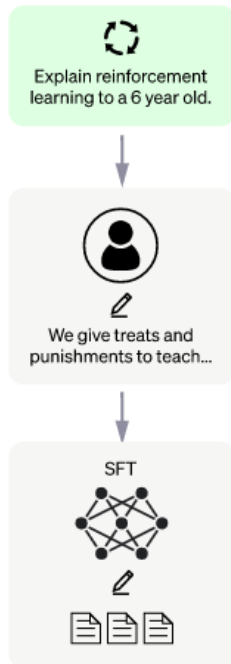
GPT-3.5

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



Step 2

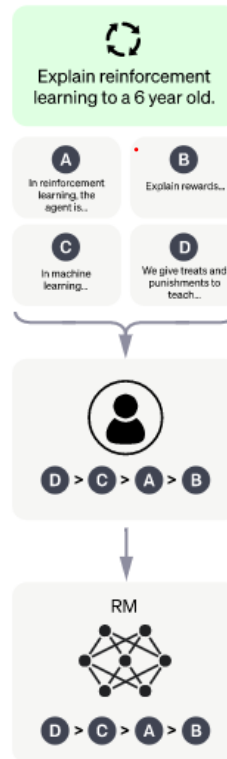
Reward Model HF

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

RL

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

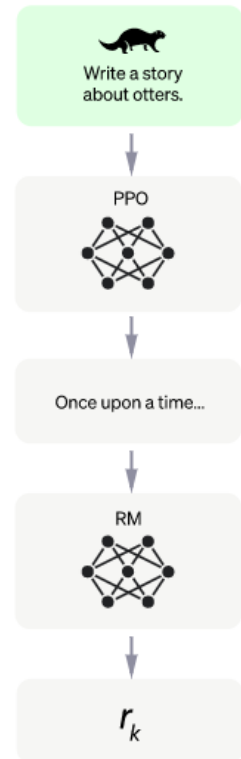
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Ethics and Fairness in AI

Agenda

What is Ethics?

Case Studies

Ethics Landscape

What is Fairness?

AI in the Future

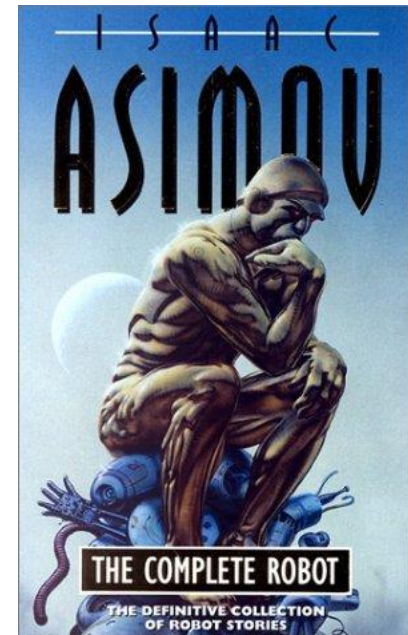
The World is Changing

- AI is finding its way into our lives
- Many of its effects go unnoticed
 - e.g., search, news recommendations, health insurance, credit score, criminal justice
- Can have unintended consequences
- We have an ethical obligation to consider the impact of our algorithms

Ethics in Artificial Intelligence

Three Laws of **Robotics**

*“A **robot** may not injure a human being or, through inaction, allow a human being to come to harm. A **robot** must obey the orders given it by human beings except where such orders would conflict with the First Law.” – Isaac Asimov (1950)*

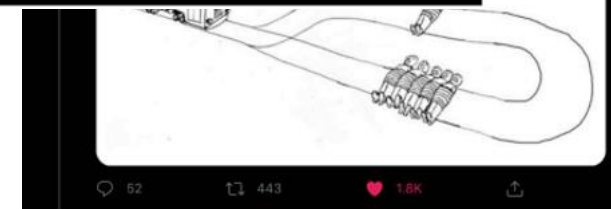
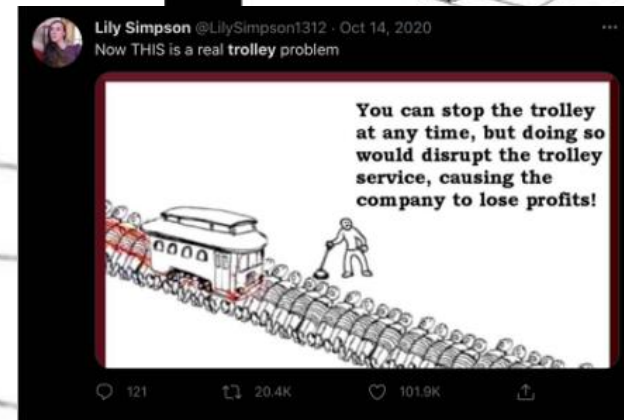
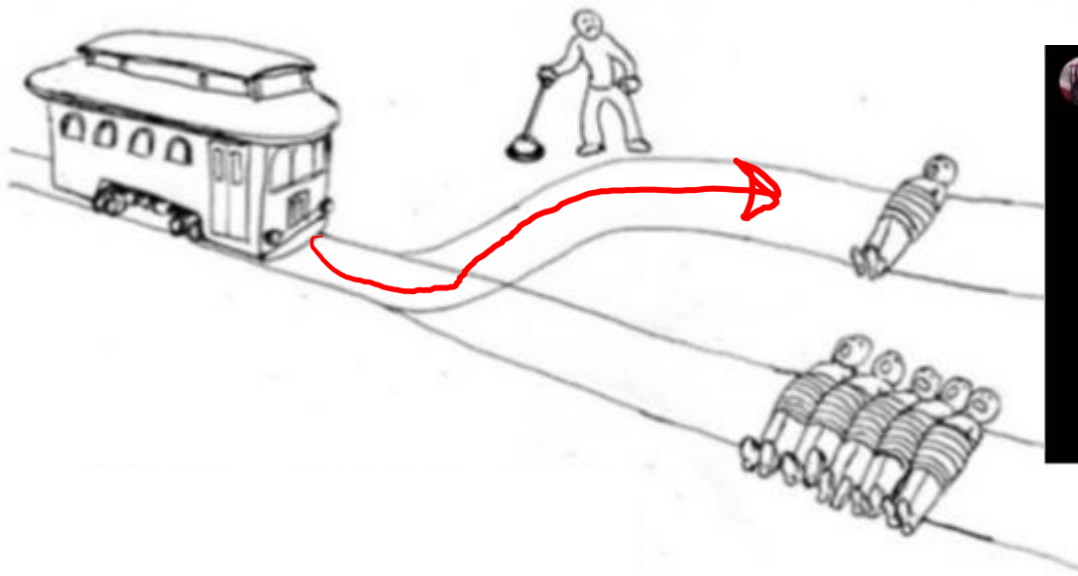


What is Ethics?

- Ethics deals with **moral principles** that govern a person's behaviour or the conducting of an activity
- **Ethics in AI** is primarily focused on social, political and cultural impact of the application of AI software

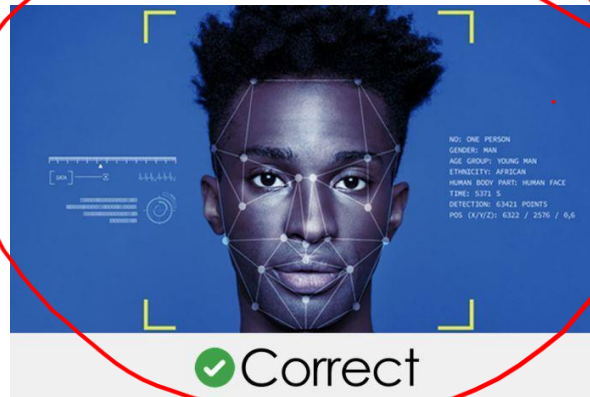
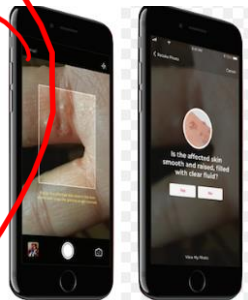
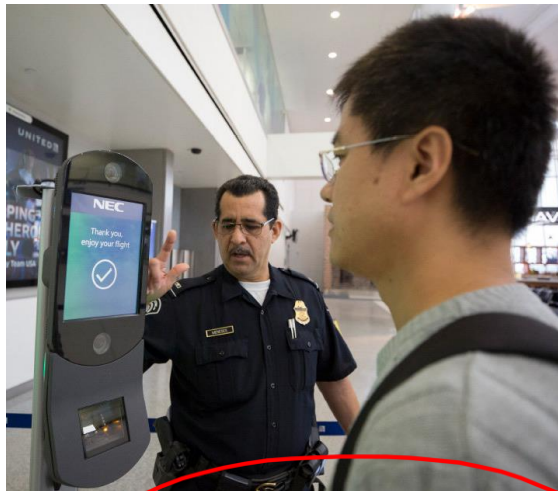
Ethics Intuition: Trolley Problem

Q: Should you pull the lever to divert the runaway trolley?



AI Ethics Landscape

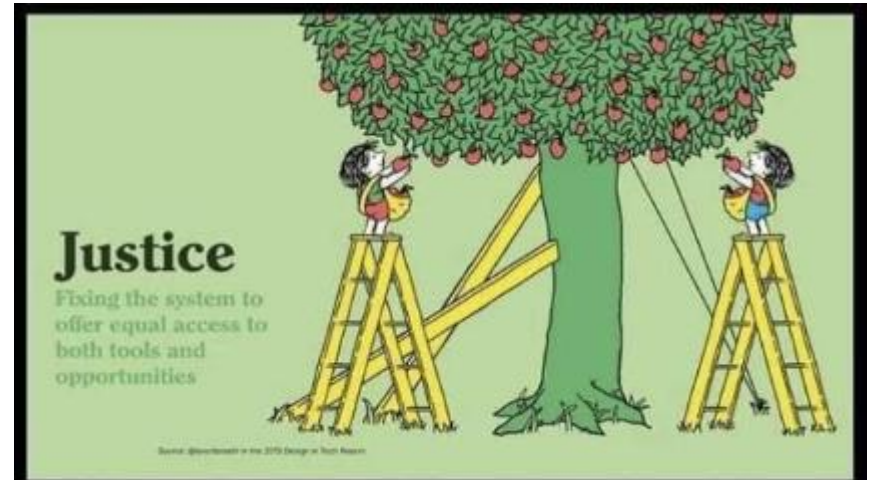
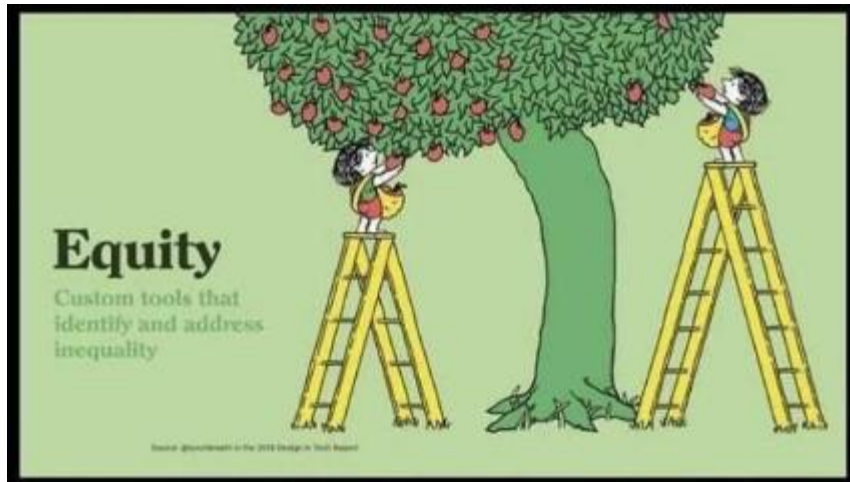
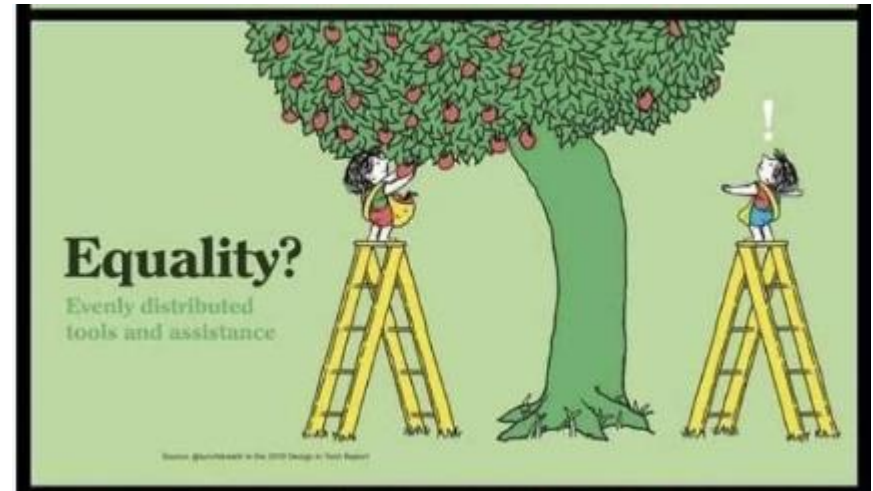
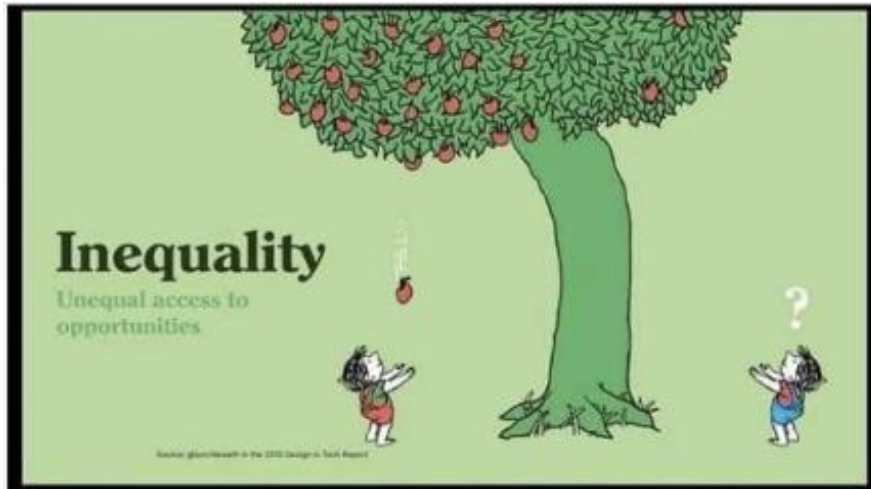
Q: What is the convergence of the ethical categories?



1. Transparency
2. Justice and Fairness
3. Non-Maleficence
4. Responsibility
5. Privacy

Fairness in Machine Learning

What is Fair?



Fairness

- What does it mean for a machine learning model to be unfair?
- Q: How can our models from the four course assignments be “unfair” or “biased”?

Let's look at some causes of bias...
... that lead to unfair models

Causes of Bias #1: Skewed Sample

- **If by chance bias is introduced, then such bias may compound over time:** future observations confirm prediction and fewer opportunity to make observations that contradict prediction.
- ex. Only crimes observed by police are recorded. Hence as more police officers are dispatched to a specific neighbourhood, then more likely that they will catch more crimes for those areas.

Causes of Bias #2: Tainted examples

- Human bias can be captured by the model.
- ex. if model uses past hiring decisions of a manager as labels, then it will learn to replicate any biases that the manager may have.

Causes of Bias #3: Limited features

- Minority groups data may have less information, or unreliably collected data.
- System tends to have much lower accuracy for the prediction of the minority group.

Causes of Bias #4: Sample Size Disparity

- When there is much less training data coming from the minority group than those from the majority group.
- Model has less information to accurately model the minority group.

Causes of Bias #5: Proxies

- Excluding sensitive attributes such as race/gender may not be enough.
- Other features such as neighbourhood could act as proxies of sensitive attribute.
- Can be difficult to decide if we should include feature or not.

Let's look at some common ways of measuring fairness...

1. Fairness as Demographic Parity

- Acceptance **rates** of applications from both groups must be equal
- $P(C=+|A=a) = P(C=+|A=b)$; $C \in \{+, -\}$, A is sensitive attribute (e.g., gender, race, etc.)
- Problem:
 - Fairness is measured at a *group* level
 - Model can hire qualified people from one group, and random people from the other

2. Fairness as Equalized Odds

- Equal true positive rate and equal false positive rate
- $P(C=+|Y=1, A=a) = P(C=+|Y=1, A=b)$
 $P(C=+|Y=0, A=a) = P(C=+|Y=0, A=b),$
Y would be whether the person is qualified
- Problem:
 - Does not help to close the gap between the two groups (e.g., not enough applicants)

3. Individual Fairness

- Similar individuals from different groups should be treated similarly
- Problem:
 - Hard to determine appropriate measure of “similarity” of inputs

Tradeoff

- **There is no consensus on how to measure fairness of a model.**
- Different measure of fairness can contradict each other!
- Optimizing fairness often means trading off accuracy

Ideas for more fair models

- **Pre-processing:** remove information correlated to sensitive attributes
- **Add regularization term:** add a “fairness” regularizer *Hard*
- **Post-processing:** change the way we use a model to make predictions

AI in the Future

Future of AI - Good or Bad?

Life 3.0: Being Human in the Age of AI

- Written by Max Tegmark, who is one of the founders of the **Future of Life Institute**
- Considers many possible futures resulting from the advancement of AI
 - altered societal structures
 - integration of humans and machines
 - friendly and dystopian scenarios
- Q: What do you think the future will look like?
Are you excited about this future?

Artificial General Intelligence

- Consider a hypothetical situation that we have developed an artificial (general) intelligence (AGI) algorithm that can replicate human-level learning.
- Q: What would be the impact or ethical implications of such an algorithm?
- Q: Does it matter if the emergence of artificial general intelligence is sudden or gradual?

Daniel Suarez TED Talk

As a novelist, Daniel Suarez spins dystopian tales of the future. But on the TEDGlobal stage, he talks us through a real-life scenario we all need to know more about: the rise of autonomous robotic weapons of war. Advanced drones, automated weapons and AI-powered intelligence-gathering tools, he suggests, could take the decision to make war out of the hands of humans. – TEDGlobal 2013

➤ TED Talk Link:

https://www.ted.com/talks/daniel_suarez_the_kill_decision_shouldn_t_belong_to_a_robot?language=en

Conclusions

“



I think people need to understand that deep learning is making a lot of things, behind-the-scenes, much better.

Geoffrey Hinton

”

Conclusions

- AI is developing rapidly
- Our legal system has not had a chance to catch up
- Government institutions are **more responsive rather than preemptive...**
can't wait for future disasters to happen
- As engineers we have a professional duty to foresee the impact of our work on the safety of the public!

Thank You!